

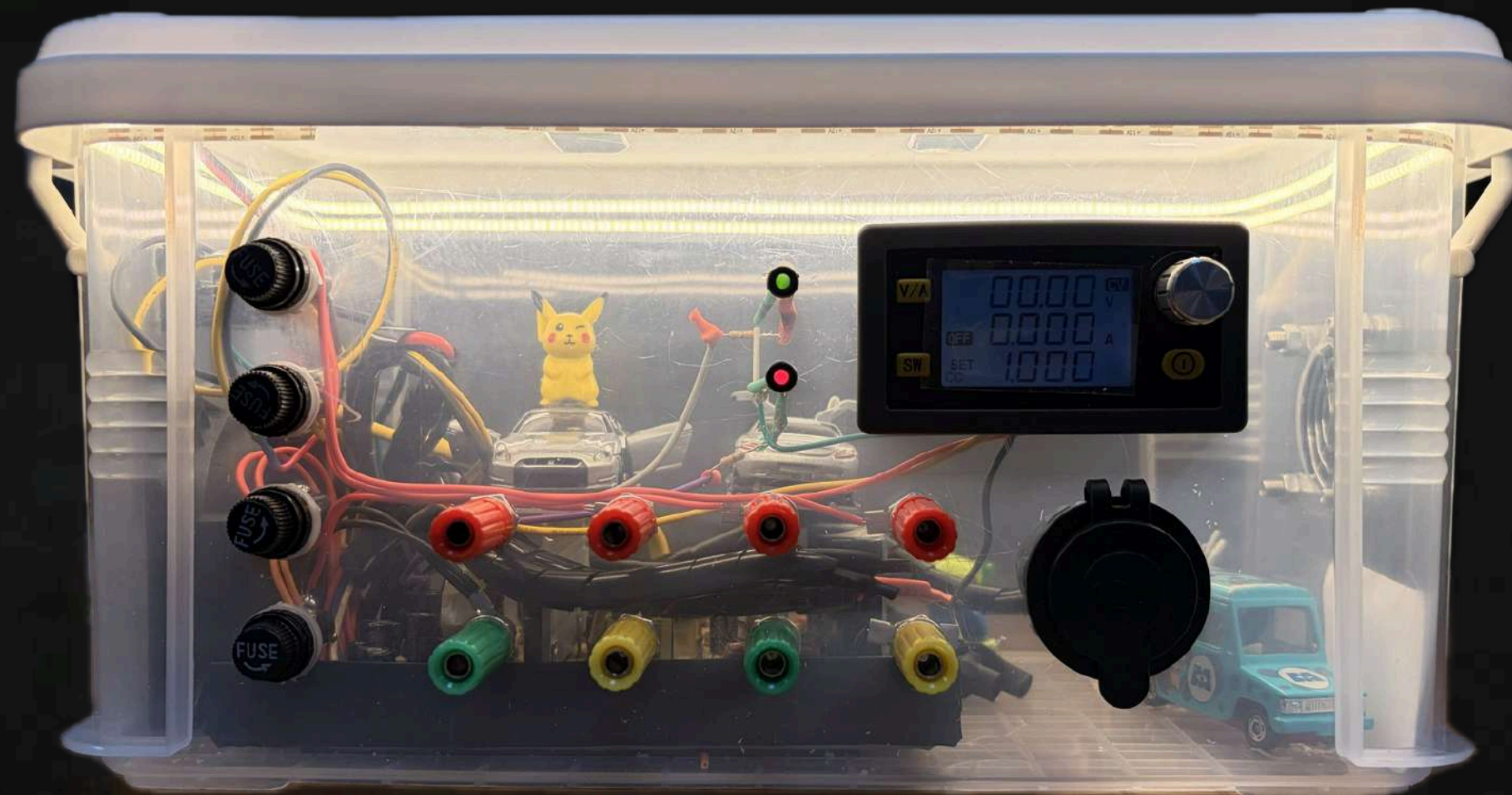
ATX BENCH POWER SUPPLY

Electronics Circuits
Laboratory Project 1

THEPNIMIT ONCHAIYA
& KORN RASARAK

OKER EB-480

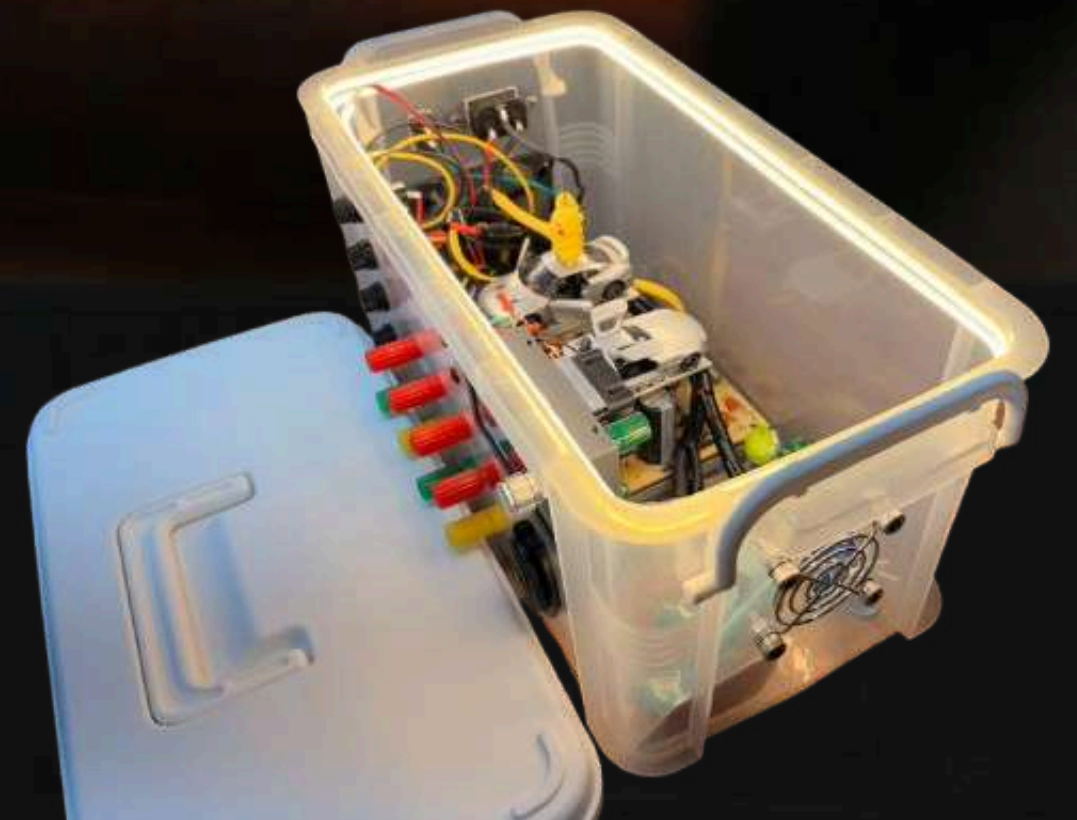
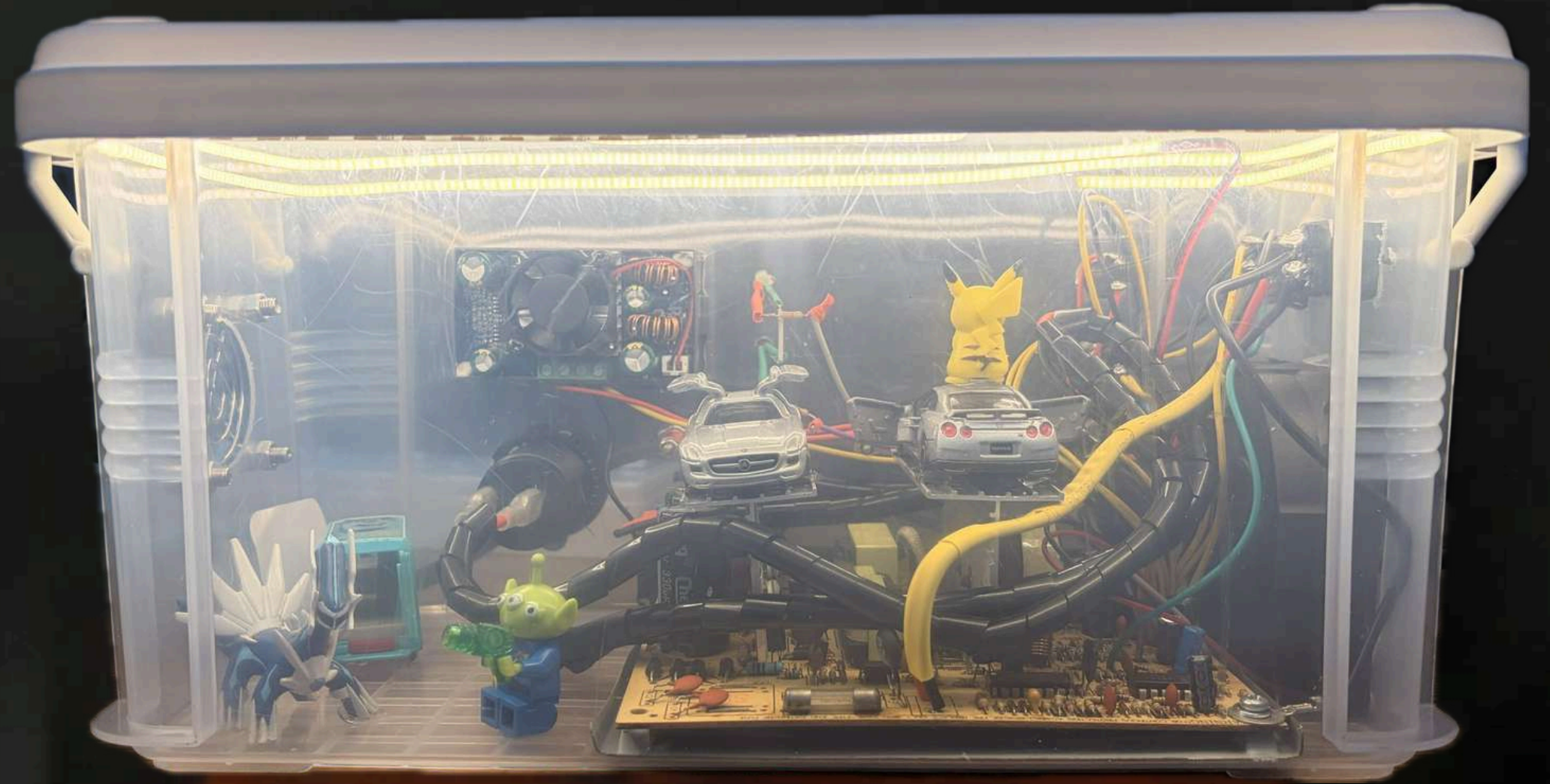
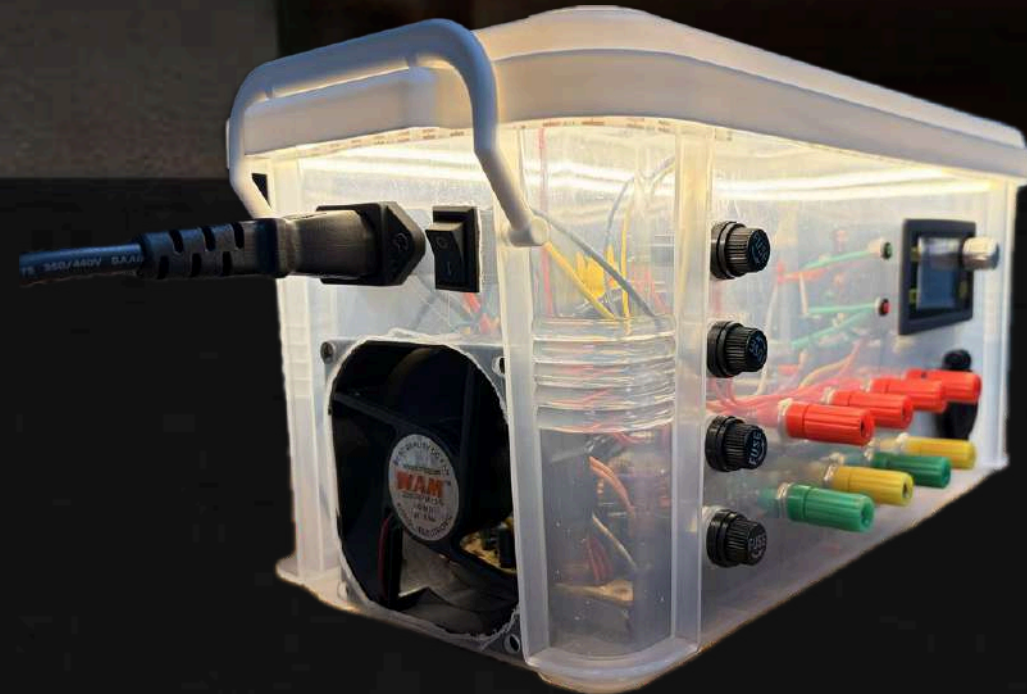
● 480W Max



PROJECT GOAL & REQUIREMENTS

REPURPOSE AN OLD ATX PSU INTO
A USEFUL BENCH POWER SUPPLY

- Reuse an existing ATX power supply
- Provide fixed +3.3 V / +5 V / +12 V outputs
- Add an adjustable output using a ZK-4KX converter
- Protect each output with a 1 A fuse
- Add clear status indication and cooling
- Build a compact and practical enclosure



SOURCE PSU & COMPONENTS

OKER EB-480

480W MAX 230V / 50HZ

ZK-4KX

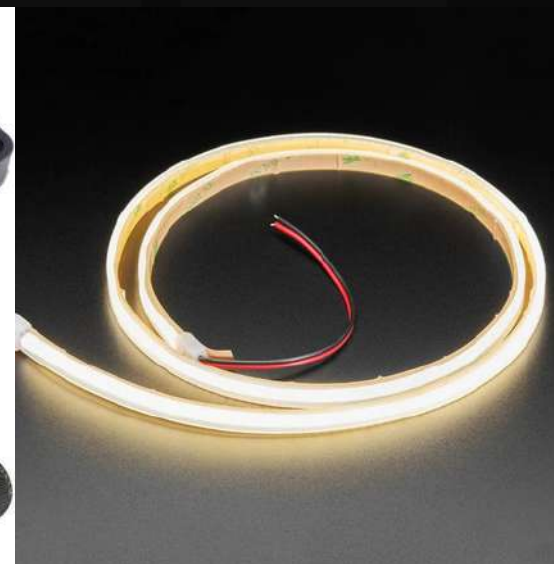
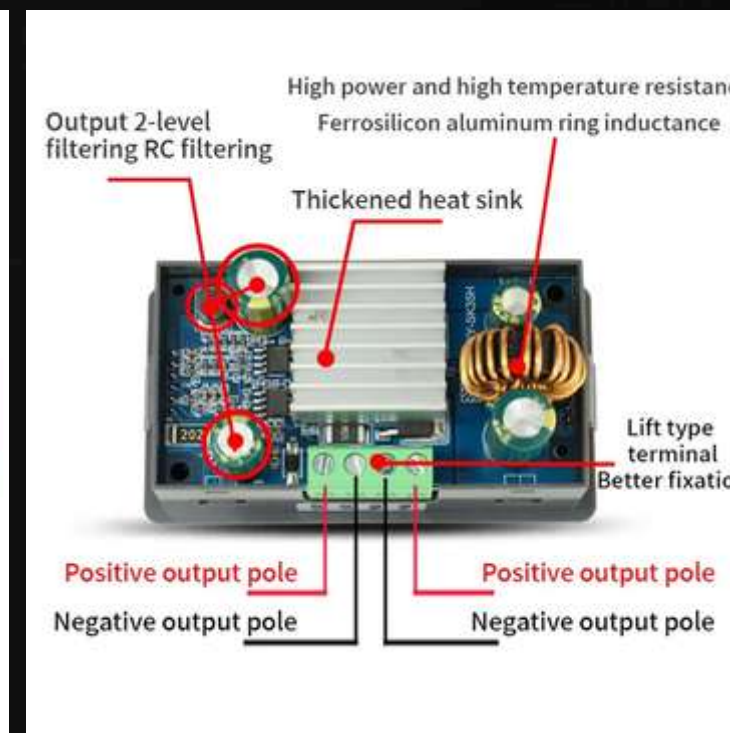
BUCK-BOOST CONVERTER

FOUR 1A 250V FUSES

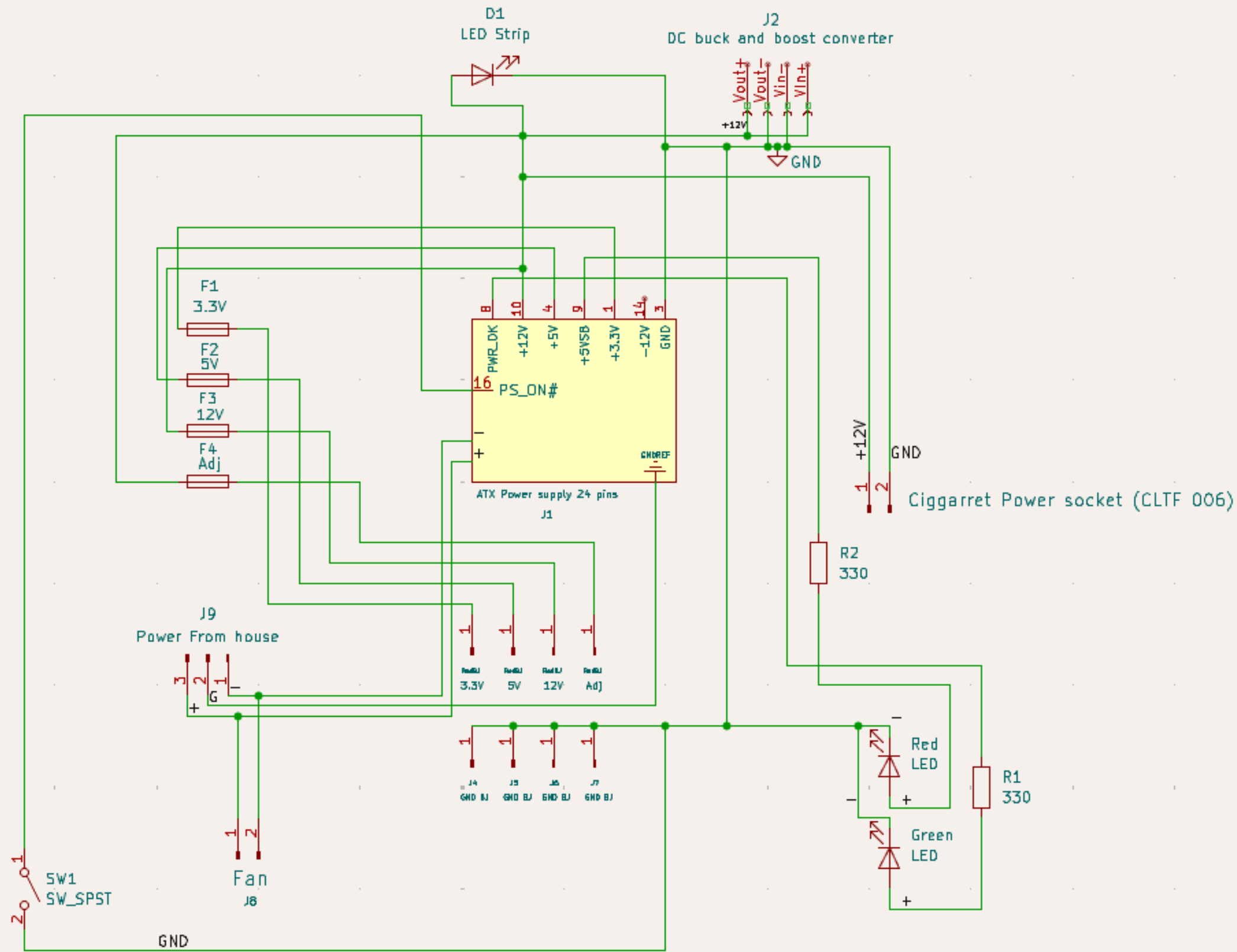
INDIVIDUAL PROTECTION

FAN, LED STRIP, DC SOCKET, BANANA TERMINALS

COOLING, POWER IN AND OUTPUTS



SYSTEM ARCHITECTURE



FIXED RAILS FOR COMMON DEVICES. **REALADJUSTABLE OUTPUT UP TO 30V.**

POWER FLOW

AC INPUT

→ **MAIN POWER SWITCH**

→ ATX POWER SUPPLY

FIXED OUTPUTS

→ +3.3 V

→ +5 V

→ +12V

→ INDIVIDUAL FUSE PROTECTION

ADJUSTABLE OUTPUT

→ +12 V INPUT

→ BUCK-BOOST CONVERTER

→ ADJUSTABLE DC OUTPUT

AUXILIARY SYSTEMS

→ **COOLING FAN**

→ 12 V LED STRIP

→ 12 V DC SOCKET

CONTROL & STATUS

→ PS_ON SWITCH

→ **STANDBY / ON LED INDICATION**

ALL CONNECTIONS ARE BASED ON THE FINAL ASSEMBLED PROTOTYPE.

ENCLOSURE & PANEL DESIGN

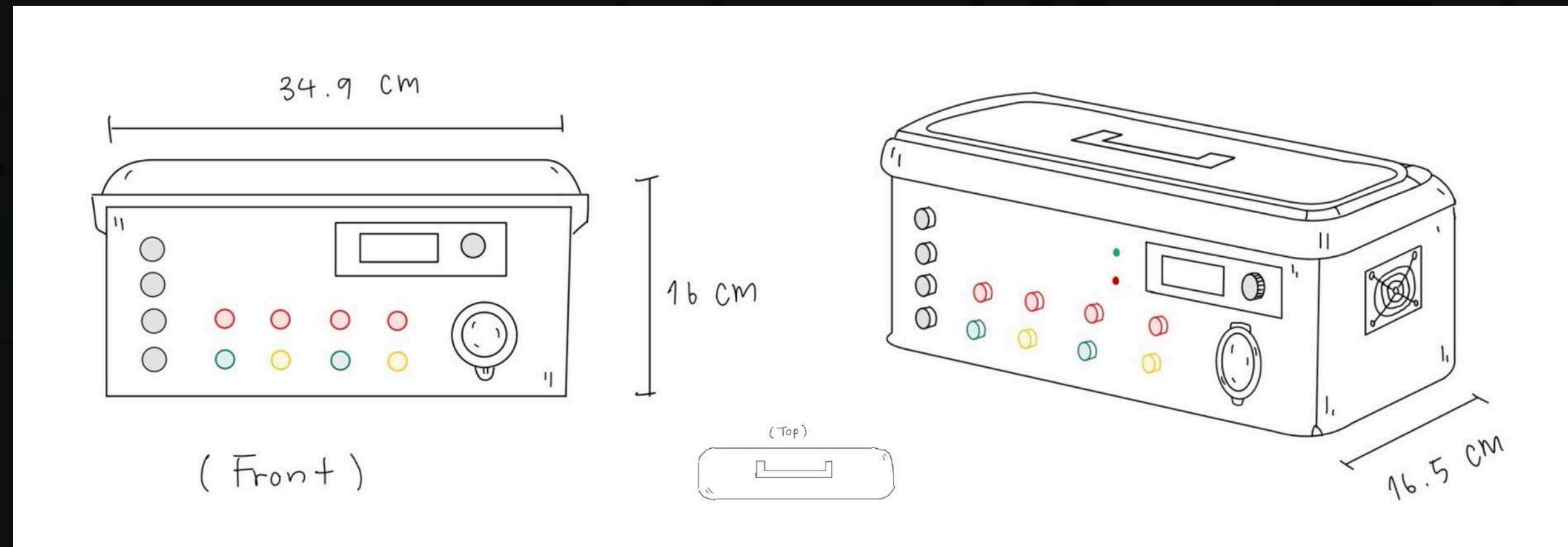
PLASTIC STORAGE-BOX ENCLOSURE

TRANSPARENT BODY FOR INTERNAL VISIBILITY

CUSTOM-MODIFIED FRONT PANEL

FUSED & LABELED OUTPUT TERMINALS

VENTILATED DESIGN FOR COOLING



FRONT VIEW



REAR VIEW



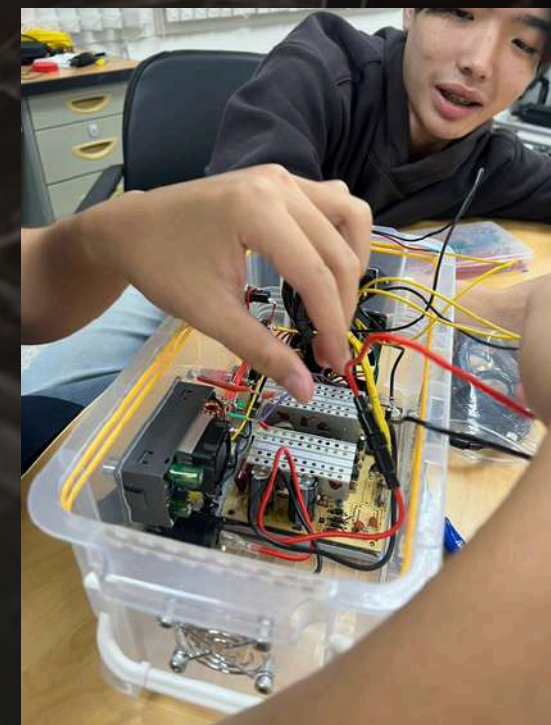
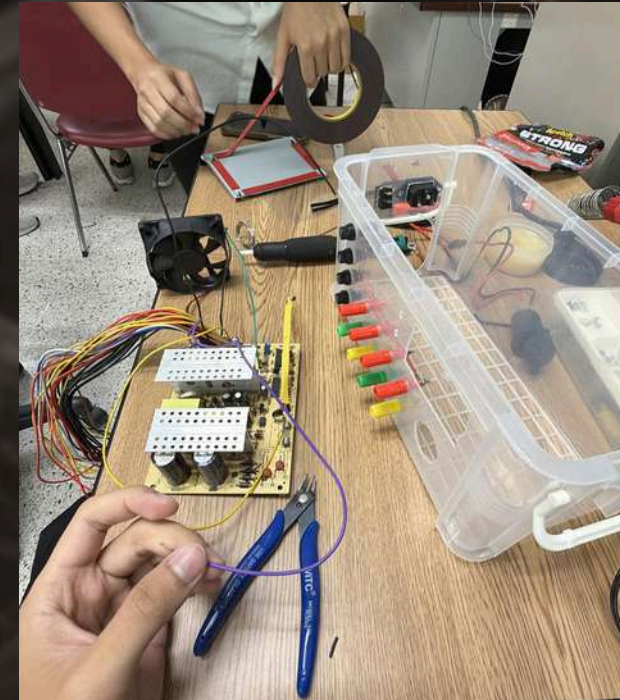
CONSTRUCTION PROCESS

Prepare ATX PSU
and components

Design & Drill the
Enclosure

Wire and Secure the
Outputs

Assemble,
Test & Inspect



MEASURED NO-LOAD RESULTS

+3.3V OUTPUT

3.44V

+5V OUTPUT

5.24V

+12V OUTPUT

11.93V

ADJUSTABLE OUTPUT

22.00V

+12V DC SOCKET

11.92V



DESIGN CALCULATIONS

FUSE PROTECTION

- 1 A FUSE IS USED FOR EACH ACCESSIBLE OUTPUT BRANCH

OUTPUT POWER

- $P = V \times I$

VOLTAGE DROP

- $V_{DROP} = V_{UPSTREAM} - V_{POST}$

VOLTAGE REGULATION

- $\% \Delta V = [(V_{LOAD} - V_{NO-LOAD}) / V_{NO-LOAD}] \times 100\%$

CONVERTER

- ADJUSTABLE OUTPUT IS SUPPLIED THROUGH THE ZK-4KX BUCK-BOOST CONVERTER

THE CALCULATIONS WERE USED TO EVALUATE OUTPUT PERFORMANCE AND PROVIDE BASIC BRANCH PROTECTION.

LOAD, RIPPLE, AND THERMAL RESULTS
NOT YET INCLUDED

SAFETY, LIMITS & TEAM

SAFETY

- DISCONNECT AC BEFORE INTERNAL WIRING CHANGES
- NEVER BYPASS OR INCREASE THE 1 A FUSE RATING
- DO NOT TOUCH INTERNAL COMPONENTS WHILE POWERED
- KEEP VENTILATION OPENINGS CLEAR

LIMITATIONS

- FIXED OUTPUTS: +3.3 V, +5 V AND +12 V
- ADJUSTABLE OUTPUT USES A BUCK-BOOST CONVERTER
- -12 V IS NOT USED IN THIS IMPLEMENTATION
- OUTPUT CURRENT MUST REMAIN WITHIN THE TESTED/APPROVED RANGE

TEAM: THEPNIMIT ONCHAIYA & KORN RASARAK

CHALLENGES & LEARNING

WIRING COMPLEXITY → LEARNED TO TRACE AND ORGANIZE POWER RAILS

LIMITED ENCLOSURE SPACE → LEARNED PRACTICAL COMPONENT PLACEMENT

HEAT MANAGEMENT → LEARNED THE IMPORTANCE OF VENTILATION AND COOLING

TESTING & TROUBLESHOOTING → LEARNED TO VERIFY OUTPUTS BEFORE OPERATION

TEAMWORK → IMPROVED COMMUNICATION AND TASK COORDINATION

